Leopard Gecko Care Guide

Eublepharis macularius

Provide real floor space, overhead warmth, low-level UVB, several hides and opportunities to explore, dig and hunt. Controlled research supports UVB-mediated vitamin D synthesis, while the exact optimal dose and long-term outcomes remain incompletely defined.

Species Profile

Common name: Leopard Gecko

Scientific name: *Eublepharis macularius*

Adult size: 7-10 in (18-25 cm) total length

Average lifespan: 15-20 years

Native range: South Asia; typical captive lines are associated most closely with Pakistani populations

Habitat and activity: Terrestrial; crepuscular/nocturnal; sheltered semi-arid rocky ground

Diet: Varied gut-loaded cultured insects

Care difficulty: Beginner to Intermediate

Shopping List

- Secure enclosure at least 36 × 18 × 18 in (91 × 46 × 46 cm; L × W × H) for one adult; see Housing for exceptions and group arrangements.
- Digital thermometers for different zones; a suitable hygrometer; timer for a consistent day/night cycle.
- Species-appropriate substrate and multiple stable hides or covered retreats.
- Reliable supply of the staple foods described below, clean feeding tools and a transport container.
- Appropriate overhead heat source if required, controlled by a compatible thermostat; infrared thermometer for surfaces.
- Appropriate visible lighting and, where recommended below, a linear UVB fixture with measured shade and exposure zones.
- Fresh-water dish or cup; insect supplements when applicable; a scale appropriate to the animal's size.

Housing

Our recommended starting adult enclosure is 36 × 18 × 18 in (91 × 46 × 46 cm; L × W × H) for one animal. Larger usable space is encouraged. These dimensions are a husbandry recommendation, not a species-specific experimentally proven threshold.

Provide multiple hides across the warm-to-cool and moist-to-drier gradient, covered travel routes and species-appropriate burrowing or low climbing structure. Fix heavy decor to the floor before adding loose substrate so digging cannot undermine it.

Juveniles may use a furnished adult enclosure when food intake, hydration and waste can be monitored. A temporary grow-out enclosure should still offer choice and cover. Quarantine newcomers separately with dedicated tools and a substrate that permits health monitoring; arrange veterinary assessment for unknown-origin animals.

Lighting & UVB

Provide roughly 10-12 hours of visible daylight followed by darkness. A measured UV Index (UVI) around 0.5-1.0 at the most exposed usable site, declining to near zero inside retreats, is our conservative starting recommendation. Low-level UVB is recommended when a usable gradient can be provided; nocturnal activity alone is not a reason to exclude light.

Measure exposure at the animal's back height with an appropriate UVI meter, such as a Solarmeter 6.5. Lamp percentage, wattage and a fixed distance alone do not establish exposure: reflectors, mesh, perch height, plant growth and lamp age change output. Keep glass/plastic out of the UV path, follow fixture safety instructions and provide complete shade. Replace or reassess lamps according to measured output and manufacturer guidance.

For albino or other light-sensitive animals, use the low end of the exposure range or lower after specialist advice, add dense shade and watch for light avoidance or eye irritation. Never increase lamp strength to compensate for uncertainty.

A controlled study found increased vitamin D status with UVB exposure even when dietary D3 was supplied. That result supports offering low UVB, but does not establish that every higher reading improves health.

Heating & Temperature

- Localized basking surface: 94-97°F (34-36°C).
- Cool-zone air: 70-77°F (21-25°C).
- Nighttime air: 65-72°F (18-22°C) for healthy established animals.

Create the gradient on one side rather than heating every hide to the same temperature. Use a compatible thermostat on each heat source; a manual lamp dimmer does not replace thermostatic protection. Secure the probe and verify temperatures independently: probe thermometers measure air/retreat temperature, while an infrared thermometer measures exposed surfaces.

Use appropriately guarded overhead warmth when needed. Keep animals clear of hot fixtures, avoid heat rocks, and switch visible lights off at night. Night heat is needed only when the room falls below the appropriate range; use a thermostat-controlled lightless source if necessary.

Humidity & Ventilation

30-50% ambient; a damp humid hide at all times. Treat readings as a starting pattern rather than a number to chase regardless of substrate, airflow or animal behavior.

Measure humidity in a representative occupied zone, away from a freshly sprayed surface. Provide sheltered moisture choices while retaining airflow. Persistent condensation, sour odor, flooded soil or surfaces that never dry indicate a setup problem. Ventilation and hydration must be adjusted together rather than simply sealing all vents.

Water & Hydration

Provide a clean, appropriately shallow dish or accessible elevated water cup at all times. Change it daily and immediately when soiled; wash it regularly with dedicated tools. Potable water is generally suitable, with treatment where local water quality requires it.

For animals that drink droplets, mist foliage or suitable surfaces at the time they become active, then allow partial drying. Droplets supplement the dish; water chemistry claims should be based on evidence rather than a species' trade name.

Substrate

For a healthy leopard gecko that has completed quarantine, a naturalistic loose substrate can be an excellent welfare choice because it permits digging and supports a more complex environment. A practical mix is approximately 60% pesticide- and fertilizer-free organic topsoil, 30% washed play sand, and 10% excavator clay or bentonite clay by volume. Mix it thoroughly, pack it several inches deep, shape stable terrain while damp, and let it dry before introducing the gecko.

Loose substrate is not appropriate for every situation. Use paper towel temporarily for quarantine, acute illness, or when stool needs close monitoring. Avoid calcium/vitamin sand, ground walnut shell, wood chips, large bark, pure coconut fiber as the only substrate, and reptile carpet. These materials either create ingestion/respiratory concerns, do not resemble the gecko's hard-packed natural ground, or can trap claws and teeth.

Impaction should not be treated as an automatic consequence of all natural substrate. Risk rises when a gecko is sick, dehydrated, kept too cool to digest normally, or offered unsuitable particulate material. Correct temperatures, hydration, feeder size, and substrate choice matter together.

Food & Feeding

Leopard geckos are insectivores. Feed a varied rotation of live, appropriately sized invertebrates rather than relying on one feeder for life. Good staples and rotation foods include dubia or discoid roaches, crickets, black soldier fly larvae, silkworms, and mealworms. Hornworms are useful for variety and hydration. Superworms and waxworms are calorie-dense and are better used sparingly, especially in adults that gain weight easily.

Hatchlings and small juveniles: small meals daily or nearly daily, adjusted to growth and body condition Older juveniles: about every other day or 4-5x/week Adults: usually 2-3x/week, adjusted to body condition, activity, season, and reproductive status Offer appropriately sized prey, remove aggressive uneaten insects, and gut-load feeders for 24-48 hours before use. Hunting is useful enrichment. Welfare studies show that leopard geckos use feeding, hiding, and climbing enrichment and prefer enriched naturalistic housing over barren setups; placing smaller prey in multiple locations can encourage hunting without increasing total calories.

Supplements

Lightly dust appropriate insect meals with calcium and use a balanced reptile multivitamin containing suitable vitamin A according to the product's directions. Gut-loading and variety support nutrition but do not eliminate the need to balance calcium-poor prey.

Match D3 supplementation to the actual UVB exposure and to vitamins already included in other products. UVB does not automatically mean zero dietary D3, and lack of UVB does not justify unlimited D3. Follow one coherent program; do not combine multiple strong vitamin products or coat every feeder heavily.

Decor & Enrichment

Provide at least three secure hides: one warm, one cool, and one humid. Add cork bark, low sturdy branches, slate ledges, textured rocks, leaf litter, and diggable areas so the enclosure offers more than empty floor space. Make all climbing features stable and low enough that falls are unlikely to cause injury. Cover along the back and sides can also help the gecko feel secure while still allowing observation from the front.

Environmental enrichment supports normal behavior. Controlled studies show leopard geckos use hides and climbing features and prefer enriched naturalistic housing. Safe textures, low climbing opportunities, multiple feeding locations, and occasional small changes to decor encourage exploration.

Handling & Behavior

Let the animal settle and feed consistently before brief optional handling. Support the body, work low over a safe surface and stop when repeated retreat, escape attempts or defensive behavior show stress. Never lift a lizard by its tail.

Avoid handling immediately after meals, during difficult sheds or when unwell. Wash hands and use dedicated equipment to reduce disease transfer.

Cleaning & Routine Maintenance

Daily: check the animal's access to water/appropriate moisture, temperatures, visible condition and equipment. Remove feces, contaminated material, dead feeders and spoiled food promptly.

Regularly: clean dishes and feeding surfaces, check thermostats and probes, inspect decor stability and maintain ventilation. Bioactive systems still require waste checks, food removal, plant care and drainage maintenance; they are not self-cleaning.

Record appetite, activity, shed history and weight or body condition when safe. Arrange veterinary assessment for unexplained changes. Quarantine new animals with dedicated tools, and disinfect only with a suitable product used at the label dilution/contact time, then rinse and dry as required before return.

Cohabitation

Individual housing is the safest routine default. Shared housing can cause food competition, injury or reproductive pressure even without obvious fighting. Never assume a male/female pair or female group is compatible. Breeding introductions belong in a planned specialist program with separate backup enclosures and individual monitoring.

Shedding & Seasonal Changes

Maintain access to a clean humid retreat and inspect toes, tail tip and eyes after shedding when observation is possible. Do not peel retained skin or apply oils/adhesives. Persistent constriction, damaged tissue or repeated failed sheds warrants veterinary help.

Seasonal changes in appetite or activity can occur. Check temperatures, hydration and weight before treating prolonged refusal as normal. Deliberate brumation or breeding cycles require healthy animals and species-specific preparation; pet care does not require forcing a seasonal shutdown.

Common Health Concerns

- Poor body condition or feeding decline: check the full gradient, hydration and food quality; arrange assessment for persistent refusal accompanied by weight loss.
- Calcium/vitamin imbalance: weakness, tremors, a soft or misshapen jaw or abnormal limbs require veterinary assessment; extra high-dose supplements are not a substitute for diagnosis.
- Skin, shed or injury problems: repeated retained skin, wounds, swelling or abnormal discoloration deserve prompt attention.
- Respiratory or systemic illness: labored/open-mouth breathing at rest, marked lethargy, loss of balance, persistent diarrhea or rapid deterioration require an experienced exotic veterinarian.

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